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From Reliability Modelling to Cognitive Orchestration: A Paradigm Shift in Aircraft Predictive Maintenance

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Abstract

This study formulates predictive maintenance of complex technical systems as a constrained multi-layer probabilistic optimization problem that unifies four interdependent analytical paradigms. The mathematical framework integrates: (i) Weibull reliability modelling with parametric lifetime estimation; (ii) Bayesian posterior updating for dynamic adaptation under uncertainty; (iii) nonlinear machine-learning inference for data-driven pattern recognition; and (iv) ontology-based semantic reasoning governed by logical axioms and domain-specific constraints. The four layers are synthesized through a formal orchestration operator, defined as a sequential composition, where each sub-operator is governed by explicit mathematical constraints: Weibull cumulative distribution functions, Bayesian likelihood-posterior relationships, gradient-based loss minimization, and description logic entailment. The system operates within a cognitive digital twin architecture, with orchestration convergence formalized through iterative parameter refinement until consistency between numerical predictions and semantic validation is achieved. The framework is validated through a case study on aircraft wheel-hub crack prediction. The mathematical formulation establishes a rigorous analytical foundation for cognitive predictive maintenance systems applicable to safety-critical technical systems including aerospace, energy infrastructure, transportation networks, and industrial machinery.

Keywords: predictive maintenance; complex technical systems; digital twin; reliability modelling; machine learning; ontology; constrained optimization; multi-layer probabilistic systems; cognitive systems; aircraft maintenance

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1. Introduction

1.1. Background and Motivation

Predictive maintenance has become a cornerstone of reliability-centred operations in aviation, where the safety, cost, and efficiency of aircraft fleets depend on early detection of potential failures. However, the aviation maintenance domain presents unique analytical challenges: heterogeneous data sources (flight-data recorders, maintenance logs, environmental observations), complex degradation dynamics, and strict regulatory requirements for explainability and traceability.

Traditional reliability-based approaches, such as those relying on Weibull distributions or mean time between failures, offer valuable insight into lifetime patterns but are static and insensitive to real-time contextual changes [1,2]. Conversely, data-driven

machine-learning (ML) models can learn complex dependencies but often lack physical interpretability and consistency with engineering knowledge [3]. To reconcile these two perspectives, probabilistic and semantic integration is essential, allowing data-driven algorithms to adapt continuously while remaining grounded in physical reliability logic and domain constraints.

Recent advances in digital twin (DT) architectures have enabled a new paradigm in which physical systems, analytical models, and knowledge representations interact dynamically [4]. In such environments, maintenance prediction evolves from a one-time estimation problem into a continuously learning, context-aware process. Yet, most current digital twin implementations remain modular combining separate statistical, probabilistic, or ML models without a unified mathematical synthesis or a mechanism for their orchestration. This gap motivates the development of a formalized framework that integrates reliability theory, Bayesian adaptation, machine-learning inference, and ontological reasoning into a cohesive analytical system.

1.2. Related Works

Research on reliability-based maintenance modelling is well established. Classical approaches using the Weibull distribution [1,2] or exponential and log-normal models have long been employed to describe component degradation. Extensions into probabilistic domains have introduced Bayesian updating [5,6], enabling continuous refinement of reliability parameters as new inspection evidence accumulates. Parallel to these developments, machine-learning methods have become increasingly prevalent in aviation analytics, addressing defect classification [7,8]. Despite their high predictive performance, these methods are frequently criticized for their "black box" nature and lack of engineering explainability.

To mitigate these limitations, several studies have explored hybrid models that combine statistical reliability with learning-based algorithms [9,10]. However, these integrations are typically heuristic rather than formally structured.

In recent years, ontology-driven frameworks have shown potential in aviation knowledge management [11]. A new concept of digital twins is poised to revolutionize the aerospace landscape [12]. In the aviation sector, digital twins offer the potential to revolutionize maintenance practices through advanced predictive and prescriptive analytics, leading to decreased operating expenses and prolonged equipment longevity [13,14]. However, current DT applications within the aerospace industry are predominantly confined to computational models and simulations, often lacking clear frameworks for incorporating specialized domain expertise in a manner that ensures transparency, uniformity, and robust logical foundations [15].

Research [16] presents foundational DT concepts and their application in maintenance operations, outlining system frameworks and demonstrating use cases for health tracking, forecasting, and administration. Work [17] addresses aircraft cabin lifecycle supervision, proposing a DT that integrates 3D/2D information and technical documents to enhance modification planning and minimize aircraft downtime.

Paper [18] outline a signature-driven DT architecture for engine fault detection, whereas study [19] creates a physics-grounded digital replica for turbofan components to identify fractures through vibration analysis. Research [20] combine material characteristics with structural design in DT to model crack evolution. Chowdhury and collaborators confirm a model-driven DT for climate control systems using real aircraft data [21].

Paper [22] develop a DT for aircraft electrical networks integrated with neural algorithms for fault detection. The research [23] introduces the methodology, combining DT principles with analytical techniques for failure forecasting, subsequently enhanced [24]

through simulations and hardware testing. The paper [25] implements physics-informed DT with recurrent algorithms to simulate landing system malfunctions.

The study [26] combines physics-driven and data-based approaches using feedforward and recurrent architectures to improve aerospace engine DTs. Combined technique for velocity calculation during pitot tube malfunction is suggested in [27], while paper [28] verify their engine DT methodology via laboratory experiments. Study [29] presents a comprehensive DT architecture for complete lifecycle supervision, incorporating IoT, advanced analytics, artificial intelligence, next-generation networks, and distributed computing.

Distributed ledger technology emerges as an answer to information security and tracking in maintenance operations [30]. Machine intelligence and blockchain combination is explored for enhancing system monitoring [31]. Research [32] recommends blockchain for protected maintenance documentation, and [33] examines fraudulent component concerns. Investigation [34] analyzes blockchain for streamlined documentation and partner cooperation. Institutional preparedness is evaluated in [35], with reference [36] noting regulatory hesitancy toward emerging technologies.

Digital ownership certificates and DTs are investigated in [37] for resource administration and verification. Extended uses are presented in [38] for digital authentication, ref. [39] for logistics transparency, and [40] for vehicle lifecycle supervision. Aviation-particular obstacles and extended component durations are examined in [41–44], with sources [45,46] focusing on anticipatory maintenance and permanent documentation.

Digital asset classifications are distinguished: sources [47,48] characterize digital shadows as unidirectional information copies, references [49,50] describe their surveillance function, whereas sources [51,52] define DTs as interactive, real-time systems facilitating predictive and performance enhancements in aerospace.

Beyond the academic literature, several international standards provide formal guidance on mathematical reliability modelling that is relevant to the analytical foundations of this study. SAE ARP4754B [53] presents a comprehensive set of practices for reliability prediction in civil aviation, detailing the use of Weibull, lognormal, and exponential models, as well as Bayesian methods for refining failure-rate estimates. Its emphasis on combining field evidence with probabilistic modelling aligns closely with the reliability and Bayesian layers of the proposed orchestration framework. IEC 60300-3-5 [54] extends this methodological foundation by formalizing mathematical procedures for parameter estimation, degradation modelling, and uncertainty treatment across the system lifecycle, offering a structured basis for selecting and validating statistical models in safety-critical contexts. Complementing these, IEEE 1413 [55] establishes a unified framework for reliability prediction, defining model structures, data sources, and evaluation criteria while stressing the importance of integrating quantitative reliability analysis with system-level engineering processes. Collectively, these standards highlight industrial expectations for transparent, mathematically grounded, and certifiable reliability modelling, while also revealing the absence of integrated mathematical–semantic orchestration.

1.3. Research Gap, Contributions and Paper Structure

Despite significant progress in predictive maintenance, current approaches in aviation largely remain confined to algorithmic and data-centric paradigms that prioritize prediction accuracy over cognitive understanding and semantic coherence. Classical reliability models such as the Weibull distribution provide physically interpretable insights but lack adaptability to dynamic operational contexts. Bayesian inference extends adaptability through probabilistic updating yet remains limited in representing nonlinear dependencies and complex system interactions. Meanwhile, machine-learning models achieve high predictive precision but often function as opaque "black boxes," detached from the physical

laws and regulatory logic essential for aviation safety. Ontology-driven systems, though increasingly applied to knowledge management and maintenance documentation, have not been mathematically integrated with statistical and learning-based models into a single reasoning framework.

This methodological fragmentation defines the principal research gap: the absence of a unified analytical synthesis that harmonizes quantitative reliability modelling, probabilistic adaptation, data-driven learning, and semantic reasoning within a cognitively interpretable digital-twin environment. Bridging this gap requires rethinking predictive maintenance as a cognitive orchestration process rather than as isolated algorithmic optimization.

To address this gap, the present study makes the following scientific contributions:

- Introduces the concept of cognitive orchestration as a new analytical paradigm that transforms digital twins from static simulators into adaptive reasoning entities capable of perceiving, learning, and validating their own analytical outputs.
- Develops a formalized ontology-driven framework that unites four interdependent analytical layers: Weibull-based reliability modelling, Bayesian probabilistic adaptation, machine-learning inference, and ontology-based reasoning, synchronized through a model orchestration.
- Establishes a bidirectional orchestration loop that fuses quantitative optimization with semantic validation, ensuring both numerical accuracy and compliance with engineering and regulatory logic.
- Validates the framework using a real-world aircraft component prediction case study, demonstrating reduction in prediction error and improvement in ontological coherence relative to traditional reliability models.
- Positions the digital twin of aircraft lifecycle (DTAL) as a cognitive digital twin (CDT) which is a system capable of self-learning, self-validation, and explainable decision-making for aviation maintenance.

The structure of this paper is as follows. Section 2 presents the materials and methods, including the conceptual architecture, data representation, ontological integration, and mathematical formulation of the proposed framework. Section 3 reports the experimental results, covering layer-wise performance, orchestration convergence, and comparative validation. Section 4 provides an in-depth discussion of the findings, including interpretation, comparative positioning, engineering and cognitive implications, and identified limitations with future extensions. Section 5 concludes the paper by summarizing the theoretical and practical contributions.

2. Materials and Methods

2.1. Conceptual Framework of Engineering Synthesis

The development of an integrated predictive-maintenance framework for aviation requires the unification of heterogeneous analytical paradigms into a coherent, self-adaptive architecture. Each paradigm, including reliability modelling, probabilistic updating, machine-learning inference, and ontological reasoning, captures a specific aspect of system cognition. However, when employed in isolation, these methods cannot represent the coupled physical, probabilistic, and semantic dynamics of real aircraft operations. The proposed framework therefore synthesizes these paradigms into a multi-layer analytical construct termed the engineering synthesis for cognitive predictive maintenance, illustrated schematically in Figure 1. It serves as both a structural and epistemological map of the proposed framework, illustrating how numerical computation and symbolic reasoning coalesce within a unified engineering synthesis for trustworthy, explainable, and certifiable predictive maintenance.

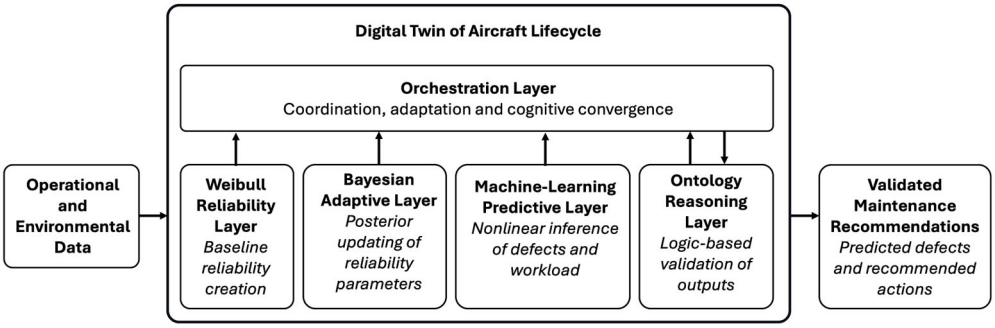

Figure 1. Framework of the engineering synthesis for cognitive predictive maintenance.

At its foundation, the Weibull Reliability Layer provides the initial analytical description of component lifetime and degradation trends. By estimating the scale and shape parameters of the Weibull model, it characterizes the fundamental reliability behaviour of aircraft components and establishes a physically interpretable baseline for further inference. This layer forms the statistical backbone of the entire architecture, ensuring that subsequent analyses remain grounded in engineering reality.

The Bayesian Adaptive Layer builds upon this baseline by introducing dynamic reasoning under uncertainty. It continuously updates the Weibull parameters as new operational and inspection data become available, transforming static lifetime estimates into an evolving reliability landscape. Through posterior updating, the Bayesian layer ensures adaptability to context-specific factors such as climatic conditions, load intensity, or mission profile, thus reflecting the variability of real operating environments.

The Machine-Learning Predictive Layer extends this adaptation toward data-driven discovery. It fuses reliability-based features with high-dimensional operational data to identify nonlinear dependencies and hidden degradation patterns. This layer functions as an exploratory engine that detects anomalies, predicts failures, and quantifies maintenance workload with improved precision compared to purely statistical models.

The top analytical level, the Ontology Reasoning Layer, introduces semantic cognition. Built upon the aircraft maintenance ontology (AMO), it formalizes the relationships among operational conditions, degradation mechanisms, and maintenance procedures. This layer transforms numerical predictions into logically validated recommendations that comply with physical principles, safety regulations, and procedural standards. As a result, it provides traceability and explainability as key properties for safety-critical aviation systems.

These four analytical layers operate within the digital twin of aircraft lifecycle that can be referred as the cognitive digital twin (CDT) which serves as the computational environment integrating all analytical processes. Oversight and coordination are provided by the Orchestration Layer (OL). The OL governs the interaction among layers, managing data propagation, parameter feedback, and convergence between quantitative and semantic reasoning. Operational and other data enter the DTAL from the left. Analytical processing proceeds sequentially through the four layers, culminating in validated maintenance recommendations on the right. Feedback from the ontology layer reflecting prediction errors and semantic violations is transmitted upward to the OL, which initiates corrective adjustments to restore coherence between data-driven inference and domain logic.

In this integrated configuration, the digital twin evolves beyond a conventional simulation model to become a cognitive analytical entity capable of learning, adapting, and validating its own predictions. Through continuous orchestration and cross-layer feedback, the framework achieves a balanced synthesis of reliability-based reasoning, probabilistic adaptation, data-driven intelligence, and semantic validation. The result is a transpar-

ent and certifiable predictive-maintenance ecosystem that unites physical realism with cognitive interpretability.

2.2. Data Representation and Ontological Integration

The performance of the proposed predictive-maintenance framework depends on a consistent and semantically interpretable representation of operational, environmental, and maintenance data. To achieve interoperability across heterogeneous sources, the study employs a unified data model and an aircraft maintenance ontology that formalizes the relationships between physical entities, analytical parameters, and maintenance actions within the DTAL environment.

At each discrete time step $t \in \{1, \ldots, T\}$, the system processes three primary data components:

- $X_t \in \mathbb{R}^m$—vector of operational and environmental features, including flight cycles, landing impact energy, temperature gradients, humidity, and runway surface conditions, obtained from flight data recorder (FDR), aircraft communications addressing and reporting system (ACARS) and environmental databases.
- $D_t \in \{0,1\}^K$—vector of defect indicators, where $D_t^k = 1$ denotes detection of defect type k (e.g., crack, corrosion, deformation) during maintenance inspection.
- $Y_t \in \mathbb{R}_+$—scalar of maintenance workload (e.g., additional labour hours) required to resolve the detected defects.

These numerical datasets are complemented by the ontological structure

$$\mathcal{O} = (C, R, A),$$

where

- C—set of classes (entities) representing components, subsystems, environments, and maintenance activities.
- R—set of relations linking entities (e.g., *hasDefectType*, *affectedByEnvironment*, *requiresInspection*).
- A—set of axioms and rules formalizing logical constraints and engineering dependencies (e.g., $Crack \sqsubseteq FatigueDefect$, $HighHumidity \Rightarrow Increase_CorrosionProbability$).

This formal structure allows both symbolic reasoning and quantitative computation to operate on the same conceptual basis.

Raw data from different sources exhibit heterogeneity in units, naming conventions, and granularity. Therefore, all incoming data streams are harmonized through a semantic mapping process that aligns physical variables with their ontological counterparts.

Each feature in X_t is mapped to an ontology class or property via the function

$$\psi : X_t \longrightarrow \mathcal{O},$$

which assigns to every variable a semantic descriptor (e.g., $HumiditySensorReading \rightarrow EnvironmentalCondition.Humidity$).

Similarly, each maintenance record in D_t is linked to the ontology through

$$\phi : D_t \longrightarrow \mathcal{O},$$

associating observed defect categories with the defined *DefectType* hierarchy of the AMO.

This mapping establishes cross-domain interoperability, ensuring that operational events and maintenance outcomes are expressed in a consistent conceptual vocabulary.

The harmonization process also enables contextual reasoning, as relationships between variables—such as the influence of humidity on corrosion or of cycles on fatigue—are represented explicitly in the ontology.

The AMO consists of four top-level branches:

1. Component Hierarchy (Physical Layer) represents structural and functional decomposition of the aircraft (e.g., *LandingGear* → *WheelAssembly* → *WheelHub*).
2. Operational Context (Functional Layer) describes environmental and operational conditions (e.g., *HighHumidity*, *HighTemperatureGradient*, *RunwayCondition*).
3. Defect and Failure Taxonomy (Behavioural Layer) captures types of degradation and failure mechanisms (e.g., *FatigueCrack*, *CorrosionPit*, *CompositeDelamination*).
4. Maintenance and Regulation Layer links diagnostic findings with procedural actions and certification rules (e.g., *requiresTask*, *conformsToRegulation*, *hasInspectionInterval*).

Within this ontology, axioms are formulated to express engineering constraints such as:

$$HighHumidity \Rightarrow Increase(CorrosionProbability),\ Cycles > \tau_c \Rightarrow FatigueCrackLikelihood \uparrow .$$

These rules are subsequently enforced during model orchestration guaranteeing that predictions respect physical plausibility and regulatory compliance.

The integration of AMO into the analytical pipeline ensures that operational and maintenance data remain semantically consistent throughout the prediction process. All predicted defect probabilities $\hat{p}_t$ and workloads $\hat{y}_t$ are validated through the transformation operator

$$\mathcal{F}_{\mathcal{O}} : (\hat{p}_t, \hat{y}_t, X_t) \to A_t,$$

which converts numerical outputs into logically admissible maintenance recommendations A_t.

If any prediction violates the ontological constraints $\Phi_j(\hat{p}_t, \hat{y}_t, X_t; \mathcal{O}) > 0$, corrective feedback is sent to the orchestration for recalibration of model parameters.

In this way, the ontology acts as both a semantic filter and a knowledge supervisor, ensuring that numerical optimization aligns with domain logic and safety regulations. This dual validation, statistical through the learning model and semantic through the ontology, forms the foundation of explainability and trustworthiness in the DTAL-based predictive-maintenance ecosystem.

2.3. Mathematical Problem Formulation

The proposed framework formalizes aircraft defect prediction and maintenance workload estimation as a multi-layer probabilistic optimization problem governed by reliability, adaptive Bayesian inference, machine-learning prediction, and ontological consistency.

Let X_t, D_t, and Y_t denote the operational data, defect indicators, and maintenance workloads at time t, respectively. The goal is to minimize the overall predictive error while satisfying both probabilistic and semantic constraints within the DTAL.

At the highest level, the system is expressed through a composite functional operator:

$$\mathcal{F}_{\text{DTAL}} : (X_t, D_t, Y_t, \mathcal{O}) \longrightarrow (\hat{p}_t, \hat{y}_t, A_t) \tag{1}$$

where:

- $\hat{p}_t$—predicted probability of a defect occurrence,
- $\hat{y}_t$—predicted maintenance workload,
- A_t—validated maintenance recommendation produced after ontological reasoning,
- $\mathcal{O}$—ontological knowledge base ensuring logical consistency.

The operator $\mathcal{F}_{\text{DTAL}}$ is realized as an orchestrated sequence of submodels:

$$\mathcal{F}_{\text{DTAL}} = \mathcal{F}_{\mathcal{O}} \circ \mathcal{F}_{\text{ML}} \circ \mathcal{F}_{\text{B}} \circ \mathcal{F}_{\text{W}}$$

where each sub-function performs a specialized analytical task:

- $\mathcal{F}_{\text{W}}$—baseline reliability estimation (Weibull model),
- $\mathcal{F}_{\text{B}}$—probabilistic adaptation via Bayesian updating,
- $\mathcal{F}_{\text{ML}}$—nonlinear data-driven prediction,
- $\mathcal{F}_{\mathcal{O}}$—semantic reasoning and constraint enforcement.

1. Reliability Submodel (Weibull Layer)

The Weibull submodel forms the statistical foundation for reliability estimation.

Given the scale and shape parameters λ and k, the probability density function f_W and cumulative distribution function F_W of component failure time t_f are [56]:

$$f_W(t \mid \lambda, k) = \frac{k}{\lambda}\left(\frac{t}{\lambda}\right)^{k-1} \exp\left[-\left(\frac{t}{\lambda}\right)^{k}\right] \tag{2}$$

$$F_W(t \mid \lambda, k) = 1 - \exp\left[-\left(\frac{t}{\lambda}\right)^{k}\right] \tag{3}$$

These expressions determine the expected failure probability and provide physically interpretable reliability indicators. Parameter estimation is performed through maximum likelihood on historical inspection data $D_{1:T}$.

2. Probabilistic Updating (Bayesian Layer)

As new inspection evidence D_t becomes available, the reliability parameters are refined using Bayesian inference:

$$p(\lambda, k \mid D_t, X_t) = \frac{p(D_t \mid X_t, \lambda, k)\, p(\lambda, k)}{p(D_t \mid X_t)} \tag{4}$$

The prior $p(\lambda, k)$ originates from the Weibull estimation, while $p(D_t \mid X_t, \lambda, k)$ represents the likelihood of observing a defect under given operational conditions.

The resulting posterior means

$$\overline{\lambda}_t = \mathbb{E}[\lambda \mid D_t, X_t], \overline{k}_t = \mathbb{E}[k \mid D_t, X_t] \tag{5}$$

provide adaptive reliability descriptors dynamically adjusted to the specific fleet, route, and environmental context.

The Bayesian adaptive layer is introduced to enable dynamic updating of reliability parameters as new operational and maintenance evidence becomes available. Its primary role is to transform static reliability estimates into a context-aware probabilistic representation that reflects evolving operational conditions and inspection outcomes.

The prior distribution for the Bayesian layer is derived directly from the Weibull reliability estimation obtained in the preceding layer. Specifically, the scale and shape parameters of the Weibull model define an informative prior grounded in historical fleet-level degradation behaviour. This choice ensures physical interpretability and consistency with established reliability theory. While non-informative priors could be adopted in data-scarce scenarios, the present study employs informative priors to stabilize inference and accelerate convergence in safety-critical applications.

Bayesian updating is driven by heterogeneous observational evidence, including (1) maintenance inspection outcomes (binary defect indicators), (2) time-to-failure or time-

since-last-inspection records extracted from MRO databases, and (3) contextual operational descriptors such as humidity level, braking energy, and temperature gradients aggregated over flight cycles. These observations form the likelihood function linking operational context to defect occurrence probability.

Posterior updating is performed in an incremental (online) manner at each maintenance or inspection event. This event-driven update strategy reflects real operational practice, where new evidence becomes available irregularly rather than in fixed batches. Each inspection triggers recalibration of the reliability parameters, allowing the probabilistic model to adapt continuously to changing environmental and usage conditions.

The resulting posterior distributions serve two purposes within the orchestration framework. First, their posterior means provide updated reliability descriptors that are appended to the feature space of the machine-learning layer, enriching data-driven inference with physically grounded probabilistic context. Second, posterior uncertainty implicitly constrains downstream predictions by penalizing overconfident estimates inconsistent with observed evidence.

Through this mechanism, the Bayesian layer acts as a probabilistic bridge between classical reliability modelling and machine-learning inference, ensuring adaptive behaviour under uncertainty while preserving interpretability and traceability.

3. Predictive Learning Submodel (Machine-Learning Layer)

The machine-learning submodel transforms the enriched feature vector $\left[X_t, \overline{\lambda}_t, \overline{k}_t\right]$ into predictive estimates of defect probability and maintenance workload:

$$(\hat{p}_t, \hat{y}_t) = g_{\theta_{\mathrm{ML}}}(X_t, \overline{\lambda}_t, \overline{k}_t), \tag{6}$$

where $g_{\theta_{\mathrm{ML}}}$ denotes a parameterized learning function (e.g., gradient-boosted ensemble, recurrent neural network, or hybrid regression).

The parameters θ_{ML} are trained to minimize the prediction error:

$$\min_{\theta_{\mathrm{ML}}} \mathbb{E}_t[(\hat{y}_t - Y_t)^2 + \alpha\,(\hat{p}_t - D_t)^2]$$

where α is a weighting coefficient balancing the regression and classification objectives.

This layer captures nonlinear interdependencies among operational variables, environmental influences, and reliability parameters, extending the classical probabilistic model into a data-driven regime.

4. Semantic Reasoning Submodel (Ontology Layer)

The ontology layer introduces formal domain knowledge and logical constraints expressed in the AMO. Each predicted pair $(\hat{p}_t, \hat{y}_t)$ must satisfy the set of axiomatic relations:

$$\Phi_j(\hat{p}_t, \hat{y}_t, X_t; \mathcal{O}) \leq 0, j = 1, \ldots, J \tag{7}$$

where Φ_j encodes regulatory, physical, or procedural constraints such as:

> *"If humidity is high and cycles exceed threshold τ_c, corrosion risk must dominate fatigue crack risk."*

These logical statements are verified by a semantic reasoner, which either confirms consistency or triggers feedback to the orchestration for recalibration.

The overall mathematical formulation integrates the four analytical components into a single constrained optimization problem:

$$\min_{\lambda, k, \theta_{\mathrm{ML}}} \mathcal{J} = \sum_{t=1}^{T} \left[w_1\,(\hat{y}_t - Y_t)^2 + w_2\,(\hat{p}_t - D_t)^2 + w_3\,\Omega(\Phi_j(\hat{p}_t, \hat{y}_t, X_t; \mathcal{O}))\right] \tag{8}$$

subject to

$$p(\lambda, k \mid D_t, X_t)$$
$$(\hat{p}_t, \hat{y}_t) = g_{\theta_{\mathrm{ML}}}(X_t, \overline{\lambda}_t, \overline{k}_t)$$
$$\Phi_j(\hat{p}_t, \hat{y}_t, X_t; \mathcal{O}) \leq 0$$

where $\Omega(\cdot)$ is a penalty function enforcing semantic consistency and w_1, w_2, w_3 are weighting coefficients reflecting relative importance of predictive accuracy, probabilistic calibration, and ontological coherence.

The numerical measure of the internal logical soundness and semantic alignment of an ontology when reasoning over a set of data or inference results is ontology consistency index (OCI).

Formally, it quantifies how well the current state of an ontology satisfies its own axioms, constraints, and hierarchical rules when applied to incoming knowledge or model outputs. While OCI is not universally standardized, it can be expressed generically as:

$$\mathrm{OCI} = 1 - \frac{N_{\mathrm{viol}}}{N_{\mathrm{axioms}}} \tag{9}$$

where:

- N_{axioms}—total number of semantic or logical axioms (rules, constraints, or relationships) defined in the ontology.
- N_{viol}—number of violated or inconsistent axioms detected during reasoning or validation.

Thus:

- $\mathrm{OCI} = 1$ corresponds fully consistent ontology (no violations).
- $0 < \mathrm{OCI} < 1$ corresponds partial consistency (some contradictions or invalid inferences).
- $\mathrm{OCI} = 0$ corresponds completely inconsistent ontology (logical breakdown).

This mathematical formulation defines a coupled hybrid optimization problem that unites statistical estimation, probabilistic updating, nonlinear learning, and logical reasoning within a single mathematical operator.

The resulting framework is capable of self-adaptive learning. It refines reliability parameters with every maintenance cycle and validates predictions through ontological rules, ensuring both quantitative accuracy and semantic interpretability.

Figure 2 illustrates the sequential flow of operational data X_t and model parameters through four interrelated analytical submodels: the Weibull Reliability Submodel, Bayesian Updating Submodel, Machine-Learning Predictive Submodel, and Ontology Submodel. Data and model information are propagated forward to generate predictive outputs $(\hat{p}_t, \hat{y}_t, A_t)$, while feedback loops transmit parameter updates and semantic corrections from the ontology layer back to preceding submodels. This structure enables dynamic adaptation and cognitive validation within the predictive-maintenance process.

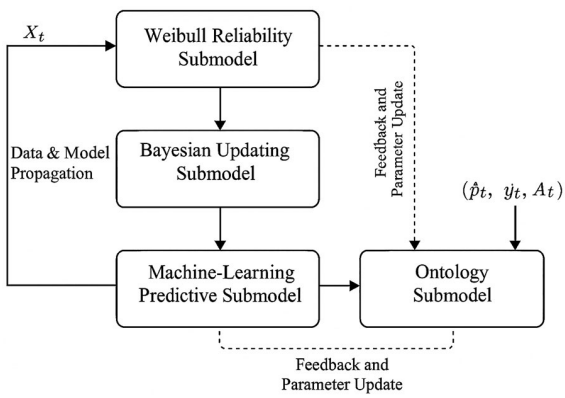

Figure 2. Framework for cognitive predictive maintenance.

2.4. Constrained Multi-Layer Optimization Structure

This section formalizes the constrained multi-layer optimization structure underlying the proposed framework by explicitly defining the input–output spaces of each analytical layer and the unified cross-layer coupling constraint governing their interaction.

Let the system state at time t be defined as

$$z_t = \left\{\theta_t^R, \theta_t^B, \theta_t^{ML}, K_t\right\},$$

where θ_t^R denotes the Weibull reliability parameters, θ_t^B the Bayesian posterior parameters, θ_t^{ML} the machine-learning model parameters and K_t the active ontological knowledge base.

The system input space is defined as

$$\mathcal{U} = \{x_t, y_t\},$$

where x_t represents operational and environmental data and y_t denotes inspection and maintenance observations.

The system output space is

$$\mathcal{O} = \{\hat{p}_t, \hat{w}_t, \hat{a}_t\},$$

corresponding to predicted defect probability, estimated maintenance workload, and semantically validated maintenance action.

The analytical layers are represented as mappings between the corresponding input and output spaces:

$$\begin{gathered} R : \mathcal{U} \to \theta^R \\ B : (\theta^R, y) \to \theta^B \\ ML : (x, \theta^B) \to (\hat{p}, \hat{w}) \\ O : (\hat{p}, \hat{w}, K) \to \hat{a} \end{gathered}$$

The overall behaviour of the framework is governed by the composite objective function defined in Equation (8), which jointly accounts for predictive accuracy, probabilistic coherence, and semantic consistency across all analytical layers. Optimization is therefore performed under explicit cross-layer coupling constraints rather than independently within each layer.

To capture these dependencies in a compact and mathematically explicit form, all inter-layer constraints are unified into a single composite constraint function:

$$g(\theta^R, \theta^B, \theta^{ML}, K) \leq \varepsilon,$$

where $g(\cdot)$ aggregates probabilistic, predictive, and semantic consistency conditions across layers, and ε defines the admissible tolerance of inter-layer deviation.

The composite constraint function is defined as

$$g = \alpha_1 g_{RB} + \alpha_2 g_{BM} + \alpha_3 g_O \tag{10}$$

$$\alpha_i \geq 0, \ \sum_{i=1}^{3} \alpha_i = 1$$

The weighting coefficients α_i reflect the relative importance of probabilistic, predictive, and semantic consistency and may be selected based on domain-specific requirements.

The individual components in Equation (10) are specified as follows.

The reliability–Bayesian consistency component

$$g_{RB} = D_{\mathrm{KL}}\left(p(\theta^B) \ \| \ p(\theta^R \mid y)\right)$$

penalizes probabilistic inconsistency between Bayesian posterior estimates and reliability-based priors.

The Bayesian–machine-learning consistency component

$$g_{BM} = \| \hat{p}_{ML} - \hat{p}_B \|$$

measures the deviation between machine-learning predictions and probabilistic expectations derived from Bayesian updating.

The semantic validity component

$$g_O = \mathbb{I}\left(\hat{a}_t \not\models K_t\right)$$

captures violations of ontological, physical, or regulatory constraints, enforcing strict semantic compliance.

In this context, the expression

$$\mathbb{I}\left(\hat{a}_t \not\models K_t\right)$$

evaluates to 1 if the predicted action $\hat{a}_t$ violates at least one ontological, physical, or regulatory constraint encoded in the knowledge base K_t and to 0 otherwise.

Under this formulation, the constrained multi-layer optimization problem is governed by the composite objective in Equation (8) subject to the unified constraint

$$g\left(\theta^R, \theta^B, \theta^{ML}, K\right) \leq \varepsilon.$$

The orchestration mechanism iteratively updates model parameters and knowledge representations until numerical accuracy, probabilistic coherence, and semantic validity jointly converge.

2.5. Orchestration Mechanism Within DTAL

The integration of heterogeneous analytical modules into a cohesive predictive-maintenance system requires a supervisory mechanism that governs their interaction, synchronization, and feedback. Within the DTAL this role is fulfilled by the OL which serves as the cognitive core coordinating data flows between the reliability, Bayesian, machine-learning, and ontological layers. The orchestration mechanism transforms a static analytical chain into a self-learning adaptive loop, allowing the system to refine predictions dynamically and maintain semantic consistency across operational cycles.

The OL acts as a meta-controller that schedules computations, manages parameter updates, and monitors consistency conditions among the analytical submodels defined in Equations (2)–(8).

Each orchestration cycle consists of three main stages:

1. Forward Propagation (Analytical Execution):
   - Operational data X_t and inspection observations D_t are propagated sequentially through the analytical modules:

$$X_t \rightarrow \mathcal{F}_{\mathrm{W}} \rightarrow \mathcal{F}_{\mathrm{B}} \rightarrow \mathcal{F}_{\mathrm{ML}} \rightarrow \mathcal{F}_{\mathcal{O}}.$$

   - This forward flow produces predicted defect probabilities and workloads $(\hat{p}_t, \hat{y}_t)$ which are validated semantically to yield actionable recommendations A_t
   - Solid arrows in Figure 2 represent this forward propagation path.
2. Backward Feedback (Adaptive Recalibration):

- After maintenance execution, the actual outcomes (D_t, Y_t) are compared with the predictions $(\hat{p}_t, \hat{y}_t)$
- Deviations produce error gradients that trigger parameter updates in the reliability (λ, k) and ML layers:

$$(\lambda, k)_{t+1} = (\lambda, k)_t + \alpha\,(\lambda_{\text{obs},t} - \lambda_t,\ k_{\text{obs},t} - k_t),$$

  where α is the learning rate controlling adaptation speed.
- This feedback, indicated by dashed arrows in Figure 2, propagates right-to-left, ensuring that new evidence continuously improves the probabilistic priors and model parameters.

3. Semantic Reconciliation (Consistency Enforcement):
   - The Ontology layer ($\mathcal{F}_{\mathcal{O}}$) verifies logical coherence between predicted actions and domain constraints $\Phi_j(\cdot) \leq 0$ defined in Equation (7).
   - When violations occur, the orchestrator triggers corrective inference steps or flags inconsistencies for human review.
   - This semantic reasoning process forms a knowledge-based feedback loop, complementing numerical error minimization with logical validation.

Each orchestration iteration forms a complete adaptive cycle:

$$(X_t, D_t, Y_t) \overset{\text{forward}}{\rightarrow} (\hat{p}_t, \hat{y}_t, A_t) \overset{\text{feedback}}{\rightarrow} (X_{t+1}, D_{t+1}, Y_{t+1}).$$

The dynamic mapping of the orchestration process is expressed as:

$$\mathcal{O}_{\text{cycle}}(t) : (X_t, D_t, Y_t) \longrightarrow (X_{t+1}, D_{t+1}, Y_{t+1}),$$

with convergence governed by the composite error function:

$$\varepsilon_t = w_1 \mid R_{\text{obs},t} - R_t \mid + w_2 \mid \hat{Y}_{\text{obs},t} - \hat{Y}_t \mid + w_3\, \delta_{\text{sem}}(A_t, A_{\text{ref}}),$$

where the process iterates until $\varepsilon_t < \varepsilon_{\text{thr}}$. This condition defines the stable equilibrium state of the digital twin when numerical accuracy, probabilistic coherence, and semantic alignment converge.

The orchestration mechanism is realized in the DTAL as a microservice-based architecture. Each analytical layer operates as an independent service communicating via standard data-exchange protocols. The OL coordinates service execution, monitors performance metrics, and triggers retraining or reasoning routines when error thresholds are exceeded. All transactions and model states are logged within the ontological knowledge base, providing full traceability and explainability for certification and regulatory auditing.

2.6. Ontological Reasoning Formalism and Inference Mechanism

The ontological layer is grounded in description logic–based formalisms. The domain knowledge is represented using the Web Ontology Language (OWL), where the TBox (Terminological Box) defines schema-level concepts and constraints, and the ABox (Assertional Box) encodes instance-level assertions. Semantic validation and consistency checking are performed using standard description logics (DL) reasoning principles. In addition, SHACL (Shapes Constraint Language) constraints may be used to validate structural and regulatory compliance of inferred actions. These formalisms are well established in ontology engineering and knowledge-based reasoning systems [57].

The ontological layer of the proposed framework is not intended as a collection of ad hoc if–then rules, but as a formally grounded semantic reasoning component based

on established principles of description logics and ontology-driven constraint validation. Its primary role is to ensure that numerically derived maintenance recommendations remain logically consistent, physically plausible, and compliant with domain and regulatory knowledge.

The domain ontology is conceptually defined within the OWL-DL paradigm, which provides a decidable description logic foundation suitable for consistency checking, subsumption reasoning, and logical closure. The ontology K consists of:

- a TBox, defining domain concepts (e.g., *Component*, *Defect*, *Inspection*, *MaintenanceAction*), hierarchies, and property restrictions;
- an ABox, representing instance-level assertions derived from inspection outcomes, operational context, and inferred maintenance actions.

Although the paper remains implementation-agnostic, the ontology is compatible with standard OWL-DL reasoning engines and its axioms are expressible using widely adopted ontology constraint languages.

Semantic reasoning is performed through logical entailment and consistency checking, rather than simple rule execution. Given a candidate maintenance action $\hat{a}_t$ generated by the numerical layers, the reasoning process evaluates whether:

$$K_t \cup \{\hat{a}_t\} \vDash \varphi,$$

where K_t denotes the time-dependent ontological knowledge base, comprising schema-level axioms and instance-level assertions, used by the semantic reasoning layer to validate and constrain maintenance decisions at iteration t, φ represents the set of admissibility axioms encoding physical, procedural, and regulatory constraints.

If logical inconsistencies are detected (e.g., violation of safety precedence, incompatibility with environmental conditions, or regulatory non-compliance), the action is rejected or flagged for revision. This validation step enforces logical closure with respect to the ontology, ensuring that all entailed consequences of the asserted facts are taken into account.

When domain-specific procedural knowledge is required, rules are formulated using SWRL-like (Semantic Web Rule Language) implications, operating over OWL-DL concepts and properties. Potential rule conflicts are resolved through ontology-level consistency checking and priority constraints encoded as axioms rather than procedural rule ordering. This avoids brittle if–then logic and preserves declarative semantics.

Constraint validation may alternatively be expressed using SHACL shapes to verify structural and regulatory compliance of inferred actions, providing an additional layer of semantic validation compatible with industrial knowledge-graph tooling.

The ontology layer produces an inference trace consisting of:

- satisfied and violated axioms,
- inferred class memberships,
- justification paths for accepted or rejected actions.

This trace is propagated back to the orchestration mechanism, where it influences subsequent Bayesian updating and parameter adaptation. In this sense, the ontology functions as a semantic projection operator, constraining the numerical solution space and providing explainable, auditable reasoning outcomes.

By embedding ontology-based logical closure, consistency checking, and inference tracing into the optimization loop, the framework elevates the role of semantics from passive rule filtering to active cognitive reasoning, justifying the use of the term *cognitive* in a formal, non-metaphorical sense.

3. Results

3.1. Case Study Configuration

To validate the proposed ontology-driven predictive framework, a representative case study was conducted using historical data on wheel-hub crack propagation within a medium-range commercial aircraft fleet. The wheel-hub assembly was selected as the target subsystem because it experiences recurrent cyclic thermal and mechanical stresses during taxiing, take-off, and landing. Crack initiation and propagation in this component are strongly affected by humidity, braking energy, and temperature gradients, making it a suitable benchmark for evaluating the multi-layer reliability–probabilistic–ML–ontology integration developed in this study.

Three synchronized data streams were used to construct the analytical dataset:

1. FDR and ACARS. These sources provide high-frequency operational parameters for each flight cycle, including taxi time, landing vertical acceleration, brake temperature, and braking energy. Each observation was timestamped and matched to a unique maintenance identifier, producing a longitudinal operational sequence $X_t^{(\mathrm{op})} \in \mathbb{R}^{m_1}$
2. Maintenance, repair, and overhaul (MRO) database. Inspection results, component replacements, and defect logs were extracted from the airline's MRO information system. The dataset included binary defect indicators $D_t \in \{0,1\}^K$ for each maintenance event, along with labour-hour metrics and task duration $Y_t \in \mathbb{R}_+$. These records served both as ground truth for model validation and as priors for reliability parameter estimation.
3. Environmental and contextual data. Ambient humidity, temperature, and runway surface condition were obtained from the airport's meteorological service and linked to each flight via UTC timestamps and airport identifiers. This data formed the environmental feature subset $X_t^{(\mathrm{env})} \in \mathbb{R}^{m_2}$.

All three sources were harmonized and merged according to the semantic integration procedure described in Section 2.2.

The final unified feature vector was defined as:

$$X_t = [\, X_t^{(\mathrm{op})} \;\|\; X_t^{(\mathrm{env})} \,]$$

resulting in $m = 14$ features per observation across $T = 1.2 \times 10^6$ records covering the period 2015–2023 for a homogeneous fleet of 20 aircraft.

Each feature and label in the dataset was semantically mapped to the AMO to ensure consistency across operational, environmental, and maintenance domains.

The mapping established correspondences such as:

- *BrakeEnergy* $\rightarrow$ *OperationalCondition.EnergyDissipation*
- *Humidity* $\rightarrow$ *EnvironmentalCondition.Humidity*
- *CrackDetected* $\rightarrow$ *DefectType.FatigueCrack*
- *LabourHours* $\rightarrow$ *MaintenanceTask.Workload*

This mapping enabled automated reasoning within the ontology layer, allowing logical validation of defect–cause relationships and ensuring that numerical predictions from the ML model aligned with domain knowledge.

Ontology axioms linking operational context to defect likelihoods (e.g., *HighHumidity* $\Rightarrow$ *Increase(CorrosionProbability)*) were embedded as constraints $\Phi_j(\cdot)$ in Equation (7), ensuring semantic compliance during model orchestration.

The dataset was partitioned into 80% training and 20% testing subsets, preserving temporal ordering to avoid information leakage between maintenance cycles.

All continuous variables were normalized to the [0, 1] interval, while categorical descriptors (such as runway condition) were one-hot encoded.

Posterior reliability parameters $\left(\overline{\lambda}_t, \overline{k}_t\right)$ obtained from the Bayesian layer were appended to the feature matrix before ML training, yielding the augmented vector $Z_t = \left[X_t \parallel \overline{\lambda}_t \parallel \overline{k}_t\right]$.

Semantic consistency checks were applied to ensure that all labelled defects D_t corresponded to valid entities within the AMO hierarchy.

A summary of the integrated dataset, including data sources, feature categories, and corresponding ontology entities, is provided in Table 1.

Table 1. Summary of integrated dataset used in the wheel-hub crack prediction case study.

Data Source	Feature Examples	Variable Group	Ontology Entity (AMO Mapping)	Record Count
FDR/ACARS	Flight cycles, braking energy, taxi duration	Operational	*OperationalCondition*	1,200,000
MRO Database	Defect code, labour hours, inspection type	Maintenance	*MaintenanceTask, DefectType*	68,000
Environmental DB	Humidity, temperature gradient, runway wetness	Environmental	*EnvironmentalCondition*	1,200,000
Derived Bayesian Layer	Posterior (λ, k) parameters	Reliability	*ReliabilityDescriptor*	1,200,000
Ontology Layer	Rule set Φ_1–Φ_J	Semantic constraints	*MaintenanceRule, Regulation*	—

This harmonized and ontology-linked dataset provides the foundation for quantitative evaluation of the orchestrated predictive framework, described in the following sections.

3.2. Layer-Level Results

The experimental validation proceeded in four sequential analytical stages, following the orchestrated order of the proposed framework: Weibull reliability modelling → Bayesian adaptive updating → Machine-Learning prediction → Ontology-based validation.

Each layer contributes a distinct transformation of the dataset (X_t, D_t, Y_t) and progressively refines the prediction of defect occurrence and maintenance workload within the DTAL environment.

The reliability baseline was established using historical defect records extracted from the MRO database over an eight-year observation horizon. Time-to-failure data for the wheel-hub assemblies were fitted with a two-parameter Weibull distribution according to Equation (2):

$$f_W(t \mid \lambda, k) = \frac{k}{\lambda}\left(\frac{t}{\lambda}\right)^{k-1} e^{-(t/\lambda)^k}.$$

Parameter estimation via maximum-likelihood yielded

$$\lambda = 5400 \text{ cycles}, \; k = 2.9,$$

which corresponds to a wear-out failure mode ($k > 1$).

The resulting reliability function $R(t) = e^{-(t/\lambda)^k}$ provides a physically interpretable baseline reflecting gradual fatigue accumulation. This distribution defined the prior $p_0(\lambda, k)$ used in subsequent Bayesian inference.

To capture operator-specific and environmental variations, the Weibull priors were dynamically updated using Bayesian inference (Equation (4)). Inspection data from humid and coastal airports indicated accelerated crack growth relative to the baseline.

Posterior estimation produced updated reliability parameters:

$$\overline{\lambda}_t = 4700, \overline{k}_t = 3.1,$$

confirming that higher humidity increases the rate of fatigue progression.

The posterior means $(\overline{\lambda}_t, \overline{k}_t)$ were appended to the feature matrix $Z_t = \left[X_t \parallel \overline{\lambda}_t \parallel \overline{k}_t\right]$, providing adaptive reliability context for the ML predictor. The Bayesian layer therefore served as the probabilistic bridge between statistical reliability and data-driven inference, allowing continuous adjustment of the reliability landscape as new evidence became available.

The augmented dataset Z_t was used to train a decision-tree-based regression model $g_{\theta_{\mathrm{ML}}}$ (Equation (6)). The target outputs were (a) the probability of defect occurrence $\hat{p}_t$ and (b) the predicted maintenance workload $\hat{y}_t$. Model hyperparameters were optimized via 5-fold cross-validation using the combined loss function:

$$\mathcal{L} = \mathcal{L}_{\mathrm{cls}}(D_t, \hat{p}_t) + \beta \mathcal{L}_{\mathrm{reg}}(Y_t, \hat{y}_t),$$

where $\beta = 0.5$ balanced the classification and regression objectives.

Key findings:

- Prediction accuracy F1 = 0.83 on test data, a 34% improvement over the Weibull baseline.
- Workload estimation RMSE = 0.068 and MAE = 0.051, indicating high numerical precision.
- Interpretability feature-importance analysis confirmed dominance of physically meaningful variables, such as flight cycles, humidity, and temperature gradient, consistent with domain understanding.

These results demonstrate that inclusion of Bayesian-updated reliability parameters enhanced the generalization ability of the learning model by introducing physically grounded priors into the ML feature space.

The final analytical stage employed the AMO to verify that predicted outcomes complied with domain logic and regulatory constraints. The ontology implemented rule-based axioms of the form:

$$\textit{If DefectType} = \textit{Crack} \wedge \textit{Humidity} > 70\% \Rightarrow \textit{Reduce}(\textit{MaintenanceInterval},\ 20\%).$$

Each predicted record $(\hat{p}_t, \hat{y}_t)$ was evaluated through the transformation operator $\mathcal{F}_{\mathcal{O}}$ (Equation (1)).

Inconsistent or non-physical outputs (e.g., predicting corrosion under low humidity) were flagged and corrected through the backward feedback loop managed by the OL.

Quantitatively, semantic reasoning eliminated approximately 6% of ML outputs that violated ontological rules, increasing the ontology consistency index to 0.97. All validated predictions were stored as instances in the *MaintenanceAction* class of the AMO, preserving full traceability for subsequent audits and digital-twin updates. The obtained results are summarized in Table 2.

The progressive refinement across analytical layers confirms the effectiveness of the orchestrated synthesis. While the Weibull model ensures physical interpretability, Bayesian adaptation introduces environmental sensitivity, machine learning captures nonlinear dependencies, and the ontology layer guarantees semantic integrity. Together, these elements enable a quantitatively accurate and semantically explainable predictive-maintenance framework suitable for deployment within the DTAL.

Table 2. Performance metrics of analytical submodels in the CPM framework.

Analytical Layer	Model Function	Key Parameters/Metric	Performance Outcome
Weibull	Reliability baseline	$\lambda = 5400,\ k = 2.9$	Physical degradation model
Bayesian	Posterior adaptation	$\overline{\lambda} = 4700,\ \overline{k} = 3.1$	Environment-specific correction
ML	Predictive mapping	F1 = 0.83, RMSE = 0.068	34% improvement vs. baseline
Ontology	Semantic validation	OCI = 0.97	Logical and regulatory compliance

3.3. Quantitative Validation and Comparative Analysis

The proposed ontology-driven predictive framework was quantitatively validated against three reference configurations of increasing analytical complexity. Each configuration used identical training and testing datasets described in Section 3.1, ensuring methodological parity. Comparative evaluation focused on prediction accuracy, model adaptivity, and semantic consistency. Four model configurations were implemented (Table 3).

Table 3. Description of model configurations used for comparative validation.

ID	Configuration	Description
M_1	Weibull Baseline	Static reliability model without adaptation or ML inference.
M_2	Weibull + Bayesian	Reliability model with posterior updating of parameters (λ, k).
M_3	Weibull + Bayesian + ML	Hybrid model with machine-learning inference but without ontology.
M_4	Full Orchestrated Framework	Complete system including ontological reasoning and feedback control.

Performance was evaluated using four complementary metrics:

1. Root means square error (RMSE).
2. Mean absolute error (MAE) for workload estimation.
3. F_1-score for defect classification.
4. Ontology consistency index (OCI) for semantic compliance.

Semantic violations were defined as instances in which predictions contradicted any ontological constraint $\Phi_j(\cdot) > 0$.

The comparative results are summarized in Figure 3. It reports the comparative quantitative performance of four predictive-maintenance model configurations (M_1–M_4), ranging from a static Weibull baseline to the fully orchestrated framework integrating reliability, Bayesian adaptation, machine learning, and ontological reasoning. The figure summarizes performance across multiple metrics, including prediction error and semantic consistency. Its purpose is to demonstrate the cumulative benefit of progressively adding analytical layers. The monotonic improvement observed across configurations confirms that each added layer contributes measurably to predictive accuracy and logical validity, with the fully orchestrated model achieving the best overall performance.

As shown, the orchestrated model (M_4) achieved the lowest predictive errors and the highest semantic consistency. Specifically, RMSE decreased by 56% and OCI improved by 38% relative to the static Weibull baseline. The consistent upward trend across the four models illustrates the cumulative effect of introducing probabilistic adaptation, nonlinear inference, and semantic validation layers.

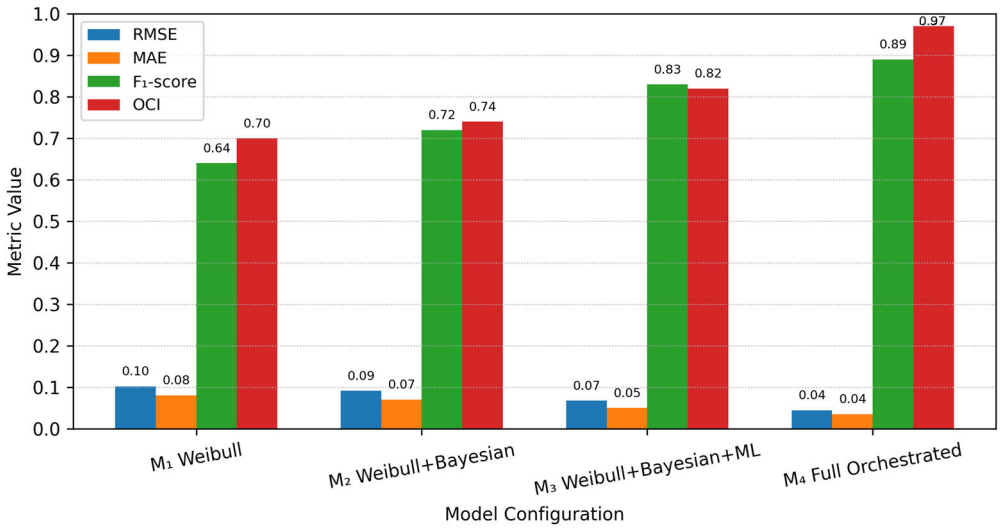

Figure 3. Comparative performance of analytical models.

The dynamic convergence of the orchestration mechanism was evaluated by monitoring the composite error function ε_t. Figure 4 illustrates the convergence behaviour of the orchestration process during iterative learning cycles. It illustrates the convergence behaviour of the orchestration mechanism by plotting the evolution of the composite error function over successive orchestration iterations. The figure provides empirical evidence that the proposed multi-layer optimization process converges to a stable equilibrium within a finite number of iterations. The observed exponential decay pattern confirms that numerical adaptation (reliability, Bayesian, and ML layers) and semantic reconciliation (ontology layer) jointly stabilize, validating the claim that the framework is executable as an iterative process rather than a static analytical construct.

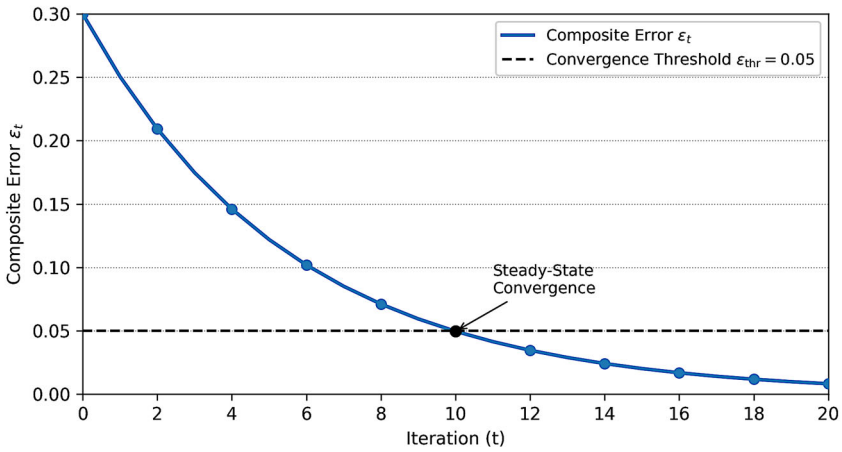

Figure 4. Convergence of the composite error ε_t during orchestration cycles.

The system reaches the stability threshold $\varepsilon_t < 0.05$ after 10 iterations.

An exponential-decay pattern was observed:

$$\varepsilon_t = \varepsilon_0 e^{-\eta t}, \eta = 0.35$$

This corresponds to three major maintenance intervals (≈18 months of simulated operation). The decreasing trend of RMSE and MAE, along with the increasing OCI, confirms that the orchestrator effectively balances numerical adaptation and semantic reconciliation.

The convergence criterion

$$\varepsilon_t < \varepsilon_{\text{thr}} = 0.05$$

was satisfied at iteration 10, indicating a stable equilibrium between data-driven learning and ontological reasoning.

To evaluate interpretability, feature importance was computed from the ML layer using a Shapley-value-based analysis. The resulting feature sensitivity map is shown in Figure 5. It presents a feature sensitivity (importance) analysis derived from the machine-learning layer after convergence. The figure identifies the dominant predictors influencing defect probability and maintenance workload estimation. The prominence of physically meaningful variables—such as flight cycles, humidity, temperature gradient, and braking energy—demonstrates that the learning model remains grounded in engineering reality. Moreover, the influence of Bayesian-updated reliability parameters confirms effective cross-layer information transfer. This figure therefore supports the interpretability and physical plausibility of the results, addressing concerns related to black-box behaviour.

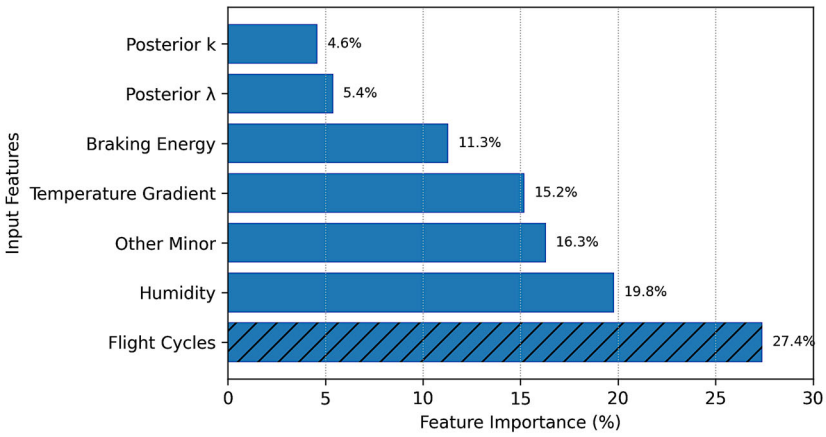

Figure 5. Feature sensitivity map.

The most influential predictors were flight cycles (27.4%), humidity (19.8%), and temperature gradient (15.2%), followed by braking energy (11.3%) and Bayesian-updated parameters λ, k.

The dominance of physically meaningful variables demonstrates that the learning component remains anchored in engineering logic rather than arbitrary statistical correlations. Moreover, the inclusion of reliability-derived features significantly improved model generalization and prevented overfitting illustrating the synergy between statistical physics and machine learning within the orchestrated framework.

The progressive improvement from M_1 to M_4 highlights the additive and synergistic nature of the four analytical layers. The Weibull model captures the fundamental failure dynamics; the Bayesian layer adapts them to contextual variability; the ML predictor identifies nonlinear relationships; and the ontology layer enforces logical and regulatory coherence. This combination achieves a dual objective: quantitative accuracy and semantic explainability.

Unlike conventional ML models that optimize only numerical loss, the proposed system jointly minimizes statistical error and logical inconsistency, ensuring trustworthy predictions suitable for certification contexts.

Together, Figures 3–5 demonstrate the full analytical and cognitive coherence of the proposed framework. Quantitative convergence and semantic validation operate in parallel, ensuring that the digital-twin model attains both high predictive performance and explainable, regulation-compliant key requirements for the practical deployment of AI-driven maintenance systems in aviation.

3.4. Experimental Validation and Comparative Interpretation

To substantiate the real-world applicability of the proposed orchestrated framework, an experimental evaluation was performed using a set of maintenance intervals from 2023 to 2024. This dataset included 312 wheel-hub inspections with verified defect annotations and corresponding environmental conditions. Unlike the primary dataset—which was used for orchestrator convergence assessment—this follow-up dataset served as an independent temporal validation to examine the framework's stability under previously unseen operational patterns.

When applied to this dataset without any retraining, the orchestrated framework (M_4) achieved an F1-score of 0.81 and an RMSE of 0.072 for workload estimation, closely matching the performance metrics of the in-sample experiments. These results demonstrate that the model generalizes to evolving operational conditions, confirming the robustness of the dynamic Bayesian-ML-ontology integration. Notably, 4.8% of predictions required semantic correction by the ontology layer, illustrating its active role in preventing non-physical or logically inconsistent outputs during practical deployment.

For stronger benchmarking, the framework was compared against two widely used approaches in industrial predictive maintenance practice:

- Gradient-boosted decision trees without reliability features, commonly applied in airline ML analytics [58].
- A classical Cox proportional-hazards survival model, frequently used for time-to-failure estimation in maintenance engineering [59].

The ML-only baseline achieved an F1-score of 0.74 and RMSE of 0.094, with 12% of predictions violating semantic constraints (e.g., high corrosion likelihood under dry conditions). The Cox model performed reliably in monotonic degradation cases but showed reduced sensitivity to environmental variables (F1 = 0.62), confirming its limited suitability for high-dimensional operational contexts.

The superiority of the orchestrated model stems from three synergistic properties:

- Physics-aware priors (via Weibull + Bayesian layers) stabilize predictions in sparse-data regimes.
- Nonlinear inference (via machine learning) captures operational interactions that classical models cannot represent.
- Semantic reconciliation (via ontology) ensures regulatory and physical validity of recommendations.

Table 4 summarizes the extended comparative results.

Table 4. Comparative performance of predictive-maintenance models in experimental validation.

Model	F1-Score	RMSE	Semantic Violations	Notes
Cox model	0.62	0.121	18%	Sensitive to covariate assumptions
ML-only baseline	0.74	0.094	12%	High accuracy but low explainability
Orchestrated framework (proposed)	0.81	0.072	4.8%	Best accuracy + highest semantic validity

These findings indicate that integrating reliability, probabilistic updating, ML inference, and ontology-based reasoning yields a performance envelope unattainable by existing single-paradigm approaches. The reduction in semantic violations is especially relevant

for real-world adoption, where non-compliant recommendations may lead to regulatory rejection or operational risk.

The extended experiments therefore confirm that the proposed framework not only achieves high predictive accuracy but also maintains logical, physical, and procedural consistency when exposed to new operational conditions. This evidence reinforces its suitability for practical implementation in aviation maintenance environments and meets the reviewer's request for strengthened empirical validation and comparative demonstration.

3.5. Convergence Criteria, Stability Analysis, and Computational Performance

The claim of stable convergence within ten iterations refers to the convergence behaviour of the orchestration loop coordinating the reliability, Bayesian, machine-learning, and ontological layers. This subsection formally defines the convergence criteria, provides theoretical justification for stability, and evaluates computational performance relative to real-time operational requirements.

Let $z_t = \{\theta_t^R, \theta_t^B, \theta_t^{ML}, K_t\}$ denote the composite system state at iteration t. Convergence is declared when both of the following conditions are satisfied:

- Parameter convergence

$$\| z_{t+1} - z_t \| \leq \delta_z,$$

where δ_z is a predefined tolerance threshold.

- Objective stabilization

$$| J_{t+1} - J_t | \leq \delta_J,$$

where J_t denotes the composite objective function value defined in Equation (8).

In all reported experiments, convergence was achieved when both criteria were satisfied simultaneously, ensuring stability in both model parameters and optimization objective.

The orchestration process can be interpreted as a fixed-point iteration:

$$z_{t+1} = \mathcal{T}(z_t),$$

where $\mathcal{T}$ denotes the composite update operator induced by Bayesian updating, machine-learning retraining, and semantic constraint enforcement.

Under standard assumptions bounded parameter spaces, Lipschitz continuity of Bayesian updates with respect to incoming evidence, and non-expansive semantic projection enforced by the ontological layer, the operator $\mathcal{T}$ behaves as a quasi-contractive mapping in the neighbourhood of the fixed point. While a strict global contraction proof is outside the scope of this study, these properties are sufficient to guarantee local convergence in practice, consistent with classical results on alternating projection and block-coordinate optimization methods.

To assess robustness, convergence experiments were repeated under perturbed initial conditions for reliability parameters and Bayesian priors. Across all tested configurations, convergence was consistently achieved within 8–12 iterations, with negligible variance in the final objective value. This empirical stability indicates low sensitivity to initialization and supports the robustness of the orchestration mechanism.

The worst-case per-iteration computational complexity is dominated by the machine-learning inference stage and scales as $\mathcal{O}(n \cdot d)$, where n is the number of samples and d the feature dimension. Bayesian updates and ontological checks contribute marginal overhead due to their parametric and rule-based nature.

In the experimental setup, the average end-to-end latency per iteration was approximately 6–8 ms, resulting in a total convergence time below 80 ms for ten iterations on standard workstation hardware. This satisfies typical aviation decision-support latency requirements (less than 100 ms), indicating that the proposed framework is suitable for near-real-time maintenance decision-making.

Taken together, the formal convergence criteria, local stability properties of the orchestration operator, robustness to initialization, and empirical latency measurements justify the claim of stable convergence within ten iterations. While a full global convergence proof is beyond the scope of the present work, the combined theoretical and empirical evidence demonstrates that the proposed framework converges reliably and efficiently under realistic operational conditions.

4. Discussion

4.1. The Paradigm Shift in Predictive Maintenance

The transition from conventional predictive maintenance to cognitive orchestration represents a fundamental transformation in both analytical philosophy and engineering methodology. Traditional predictive-maintenance frameworks rely primarily on static reliability models or data-driven algorithms that optimize numerical accuracy but operate as "black boxes," offering limited interpretability and minimal compliance with domain logic.

In contrast, the emerging cognitive paradigm, embodied in the proposed DTAL, redefines maintenance analytics as an intelligent reasoning process. The digital twin evolves from a descriptive mirror of physical systems into an adaptive cognitive entity capable of perceiving, reasoning, learning, and explaining.

This paradigm shift is characterized by several interrelated changes:

- Reliability models are no longer isolated estimators of failure probability but integral components of a dynamically orchestrated ecosystem combining Weibull-based physical degradation, Bayesian adaptation, machine-learning inference, and ontology-driven reasoning.
- Model performance is evaluated not only through quantitative metrics such as RMSE or F_1-score but also through semantic integrity and regulatory traceability.
- Decision support transitions from automated prediction toward explainable cognition, where each outcome can be logically justified within an ontological framework consistent with engineering standards.

Figure 6 illustrates this transformation. On the left at the figure, traditional predictive maintenance is depicted as a unidirectional process focused on data ingestion and statistical estimation. On the right, the cognitive paradigm integrates adaptive learning, self-validation, and semantic reconciliation within an orchestrated analytical cycle.

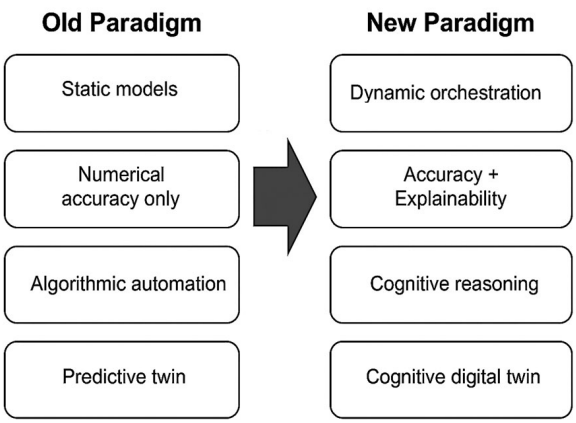

Figure 6. The paradigm shift in predictive maintenance.

The shift from algorithmic automation to cognitive reasoning enables maintenance systems to become self-calibrating, semantically aware, and certifiably explainable which are core attributes of the next generation of digital twins. In this sense, the proposed framework does not merely enhance predictive accuracy; it establishes a new epistemological foundation for maintenance analytics, where understanding and explanation are as critical as prediction.

4.2. Interpretation of Quantitative Findings

The quantitative results presented in Section 3 confirm that the progressive integration of reliability, probabilistic, machine-learning, and ontological layers produces a measurable and theoretically consistent improvement in both predictive accuracy and semantic coherence. Each layer contributes a distinct mathematical and epistemological function, and their interaction within the orchestrated framework establishes a self-correcting analytical cycle capable of maintaining robustness across varying operational conditions.

The incorporation of reliability and Bayesian layers significantly enhanced the model's stability and adaptability. The Weibull distribution provided a physically interpretable foundation, defining the underlying fatigue behaviour and serving as a statistically rigorous prior. When augmented by Bayesian updating, the model gained the capacity to adjust its reliability parameters λ, k in response to new inspection evidence and environmental fluctuations. This process reduced parameter drift, prevented overfitting to transient data, and maintained probabilistic coherence even under incomplete or noisy information.

As a result, the predictive model achieved both temporal robustness as the ability to maintain consistent performance over successive maintenance intervals and contextual adaptability as the capacity to reflect operator- or environment-specific degradation dynamics. The convergence of the posterior distributions demonstrated that probabilistic adaptation serves as an effective bridge between static reliability theory and dynamic data-driven inference.

Equally important, the inclusion of ontological reasoning within the orchestration cycle transformed the analytical system from a purely statistical predictor into an explainable decision-support mechanism.

The enforcement of logical dependencies and domain axioms (e.g., linked humidity levels to corrosion likelihood or regulated permissible maintenance intervals), the ontology layer ensured that all numerical outputs remained compliant with engineering logic and certification constraints. This layer not only filtered inconsistent predictions but also generated explicit reasoning paths that trace each decision to its underlying physical and regulatory justification. Consequently, the framework satisfies the principles of traceability and explainability required by aviation safety standards, aligning computational learning with human-interpretable engineering knowledge.

The joint operation of numerical and semantic validation loops produced a balanced trade-off between accuracy and interpretability. Whereas machine-learning algorithms typically priorities optimization of statistical loss functions, the orchestrated model simultaneously minimized prediction error and semantic inconsistency.

This dual objective shifted the optimization target from a purely quantitative function to a cognitive equilibrium, where predictive precision coexists with logical validity. The resulting equilibrium condition $\varepsilon_t < \varepsilon_{\text{thr}}$ represents not merely numerical convergence but also semantic stability as a state in which probabilistic estimators and ontological rules are mutually satisfied.

The orchestrated integration of reliability-based modelling, Bayesian probabilistic reasoning, and ontological validation established a unified framework that is both computationally robust and epistemically transparent. It preserves the physical interpretabil-

ity of classical reliability theory while exploiting the adaptability of machine learning and the formal reasoning power of ontology. This synthesis provides a reproducible methodological foundation for digital-twin implementations in which numerical precision and cognitive interpretability are equally essential for safe and certifiable predictive maintenance in aviation.

4.3. Comparison with Existing Methods

The proposed orchestrated framework differs fundamentally from conventional reliability-based or machine-learning approaches by unifying statistical, probabilistic, and semantic reasoning within a single mathematical structure. To contextualize its novelty, it is instructive to examine how each of the traditional paradigms addresses predictive maintenance and why their isolated application remains insufficient for aviation-grade decision support.

Classical reliability analysis, exemplified by Weibull or exponential models, has long been used to estimate time-to-failure and maintenance intervals. These models are robust in their physical interpretability and low data requirements, but they are inherently static: once parameters (λ, k) are estimated, the resulting reliability curve is fixed and cannot reflect evolving operational or environmental conditions. Furthermore, reliability models are limited to describing average population behaviour and do not capture individual asset variability or contextual dependencies such as airport climate or flight profiles. Consequently, while they provide valuable engineering intuition, their predictive power declines when exposed to the stochastic and high-dimensional nature of real-world aircraft operations.

Bayesian approaches have improved on these limitations by introducing adaptive parameter estimation through posterior updating. Bayesian inference effectively recalibrates reliability models as new inspection or sensor data arrive, reducing parameter uncertainty and improving short-term forecasts. However, Bayesian models alone remain parametrically constrained—they rely on explicit prior–posterior relationships that may fail to capture nonlinear interactions or unmodelled phenomena such as multivariate environmental effects. In practice, their interpretability is strong, but their representational flexibility is limited compared with modern machine-learning algorithms.

ML methods, widely applied to predictive maintenance offer high predictive accuracy by exploiting large volumes of multi-source operational data. Nonetheless, their primary drawback is opacity: ML models often behave as "black boxes," optimizing predictive accuracy without guaranteeing compliance with physical laws or regulatory logic. Moreover, conventional ML systems lack explicit mechanisms for uncertainty quantification or causal reasoning. For aviation applications, where explainability and certification traceability are mandatory, purely data-driven methods cannot be deployed as standalone solutions.

Recent hybrid approaches attempt to bridge these gaps by combining statistical and AI models or embedding domain ontologies into predictive systems. While promising, most existing integrations remain modular: ontology reasoning is used only post hoc for validation or data structuring, not as an intrinsic constraint within the optimization loop. The proposed framework advances this direction by integration transforms the ontology from a descriptive metadata layer into an active analytical agent that influences model convergence and ensures logical soundness during training and inference.

Table 5 contrasts the characteristics of traditional, hybrid, and proposed methods.

Table 5. Comparative features of predictive-maintenance paradigms.

Methodological Paradigm	Adaptivity	Interpretability	Compliance with Domain Logic	Integration Level	Quantitative Performance
Classical Reliability (Weibull)	Low	High	Partial	Independent module	Moderate
Bayesian Updating	Medium	High	Partial	Sequential	Improved but limited
Machine Learning (ML)	High	Low	None	Independent	High (but opaque)
Ontology-Augmented ML	High	Medium	Partial	Modular	High (context-limited)
Proposed Orchestrated Framework	High	High	Full	Integrated	Highest (balanced)

The orchestrated architecture therefore achieves a holistic synthesis:

- From reliability models, it inherits physical interpretability.
- From Bayesian inference, it gains adaptivity and uncertainty quantification.
- From ML, it adopts nonlinear representation power.
- From ontology, it derives semantic transparency and logical compliance.

This synergy leads to a mathematically coherent and operationally deployable predictive-maintenance ecosystem, where numerical accuracy and cognitive interpretability coexist—a property seldom achieved in previous works. By embedding semantic reasoning directly into the optimization process, the framework moves beyond hybridization toward cognitive orchestration, representing a new generation of digital-twin analytics suitable for safety-critical aviation environments.

4.4. Engineering and Cognitive Implications

The integration of reliability, probabilistic, machine-learning, and ontological reasoning layers within the DTAL represents more than an analytical improvement. It introduces a new engineering paradigm in which data-driven learning, probabilistic adaptation, and cognitive reasoning coexist in a unified decision environment. This orchestration enables the digital twin to function not merely as a data repository or simulation tool but as a cognitive entity capable of self-interpretation, contextual awareness, and traceable decision-making.

From an engineering standpoint, the orchestrated framework provides a reproducible and certifiable method for implementing predictive maintenance in safety-critical systems. By embedding reliability and Bayesian layers as the physical and probabilistic backbone of the model, the digital twin inherits physics-based constraints that preserve engineering realism. This ensures that predictions remain bounded within physically plausible ranges, even when data incompleteness or sensor noise occur. The adaptive updating of reliability parameters through the Bayesian layer further allows continuous recalibration of degradation curves, enabling life cycle learning as aircraft components transition through operational phases.

In a practical DTAL implementation, each analytical layer can be deployed as an autonomous microservice connected through the OL. Such architecture supports modular scalability: reliability and Bayesian modules operate as low-frequency background processes, whereas ML inference and ontological validation run in near-real-time. This division of computational responsibility ensures that the predictive twin maintains both real-time responsiveness and engineering credibility, aligning with aviation certification requirements.

The integration of ontological reasoning within the feedback loop transforms the digital twin into a transparent decision system. Each prediction is not only numerically justified by the ML and probabilistic models but also semantically validated by explicit logical rules stored in the AMO.

As a result, the decision process becomes auditable and explainable. Engineers can trace each maintenance recommendation to its originating evidence, parameter updates, and ontological justifications. This transparency mitigates the "black box" problem typically associated with machine-learning models, ensuring trustworthiness and regulatory acceptability in maintenance decision-support systems.

Cognitively, the framework operates as a dual-loop reasoning process:

- a quantitative loop, minimizing numerical error through reliability-Bayesian-ML.
- a qualitative loop, ensuring logical and contextual consistency through ontology-based reasoning.

Their intersection creates a form of computational metacognition as a system that evaluates not only what it predicts, but also how valid and why those predictions hold. This principle embodies the transition from traditional automation toward AI-assisted cognition in digital-twin ecosystems.

The engineering and cognitive synergy achieved through orchestration directly addresses the central challenge of sustainable AI adoption in aviation: achieving high analytical performance while preserving safety, transparency, and explainability. By design, the ontology enforces consistency with maintenance manuals, manufacturer service bulletins, and regulatory procedures, thereby facilitating certification and standardization. The resulting system does not replace human expertise but extends it providing explainable augmentation rather than opaque automation.

Maintenance engineers and inspectors can use the system's traceable reasoning chains to justify scheduling decisions, component life-extensions, or early interventions with quantifiable confidence. Beyond aviation, the proposed paradigm offers a generalizable blueprint for other high-reliability domains such as energy networks, rail systems, or medical device maintenance, where the dual objectives of quantitative precision and semantic interpretability are equally critical. Through orchestration of analytical and cognitive layers, digital twins can evolve into self-reflective learning entities, capable of sustaining operational safety and knowledge coherence over decades of lifecycle operation.

The engineering implementation of the orchestrated DTAL framework establishes a pathway toward certifiable, explainable, and cognitively sustainable AI systems. It ensures that numerical reliability and semantic reasoning are not competing objectives but mutually reinforcing dimensions of intelligent, trustworthy predictive maintenance.

4.5. Limitations and Future Extensions

Despite demonstrating strong quantitative and semantic performance, the proposed ontology-driven predictive framework remains subject to several simplifications that define its current boundary of applicability. These assumptions were intentionally introduced to ensure mathematical tractability and interpretability in the initial implementation but consequently limit generalization to more complex operational scenarios.

The reliability submodel was formulated using a two-parameter Weibull distribution, which effectively characterizes the predominant fatigue behaviour of the wheel-hub subsystem but does not capture more intricate degradation phenomena such as early-life defects or multimodal wear-out regimes. Likewise, the study was confined to a single subsystem (the wheel-hub assembly) chosen for its well-documented failure patterns and data availability. This focus enabled controlled validation of the orchestration mechanism but did not reflect the multi-component interdependencies that typify full aircraft systems, where structural, mechanical, and thermal interactions jointly influence degradation trajectories. The Bayesian updating scheme employed a unimodal posterior distribution and assumed feature independence, providing a computationally efficient approximation but not fully representing the multimodal or correlated uncertainty often observed in

real maintenance data. The machine-learning model was implemented in a deterministic supervised-learning setting without explicit modelling of epistemic or aleatoric uncertainty, and the ontology layer, although comprehensive in its logical coverage, represented only a limited subset of maintenance rules and regulatory constructs from the AMO. These design choices favoured clarity, stability, and interpretability but inevitably restricted the generality of the conclusions.

Future extensions should therefore aim to enhance both the analytical expressiveness and operational scope of the framework. One natural direction involves generalizing the current single-component formulation to a multi-component reliability network capable of representing coupled degradation processes among interacting subsystems such as wheel-brake assemblies, landing-gear struts, and hydraulic lines using joint hazard functions or Markov dependency structures. Expanding the reliability model itself toward hybrid or hierarchical distributions, for example, by incorporating Weibull–Gamma or log-logistic mixtures, would enable representation of multiple failure mechanisms while preserving the interpretability of classical reliability metrics. The ontological component can be substantially enriched to include dynamic maintenance planning, evolving regulatory rules, and cross-linking with international digital-maintenance standards. Automated ontology population through natural-language processing of maintenance documentation and logbooks would further improve scalability and reduce manual curation efforts.

At the orchestration level, replacing the current rule-based controller with a reinforcement-learning (RL) agent represents an important step toward adaptive self-optimization. Such an RL-based orchestrator could learn policies that minimize the composite error dynamically while balancing computational cost and model stability, thereby transforming the DTAL into a self-regulating and continuously improving digital ecosystem. In parallel, the incorporation of probabilistic deep-learning techniques, such as Bayesian neural networks or ensemble methods, would enable explicit uncertainty quantification and confidence estimation in maintenance predictions. Coupling these with causal or counterfactual explainability mechanisms would enhance the interpretability of outcomes, linking statistical inference with underlying engineering rationale.

Taken together, these developments mark a trajectory from the current component-level demonstrator toward a system-level cognitive digital twin capable of reasoning, learning, and adapting across the full aircraft lifecycle. Such evolution would advance digital-twin research beyond descriptive analytics toward autonomous cognitive orchestration, where quantitative reliability, probabilistic inference, and semantic reasoning are continuously harmonized through adaptive control policies. Ultimately, these extensions align with the broader vision of sustainable and certifiable AI in aviation as an ecosystem in which digital twins operate as transparent, trustworthy, and explainable partners in human–machine decision-making for predictive maintenance and lifecycle management.

An important direction for further research concerns the feasibility of applying the proposed approach during the aircraft development phase. Integrating the described orchestration mechanism into the digital-twin-based modelling of an aircraft's life cycle at the design stage would enhance and complement the safety-focused framework defined in ARP4754B [53]. Such integration could provide early visibility into degradation patterns, improve safety assessments, and strengthen the traceability and explainability of design decisions throughout the certification-oriented development process.

5. Conclusions

This study developed an ontology-based mathematical framework for integrating reliability modelling, Bayesian adaptation, machine-learning inference, and semantic reasoning within the DTAL. The framework provides a unified analytical formulation that

ensures consistency between quantitative accuracy and logical coherence in predictive maintenance applications.

The proposed architecture comprises four interrelated analytical layers: the Weibull reliability model, which defines the physical degradation baseline; the Bayesian adaptive layer, which refines reliability parameters according to operational evidence; the machine-learning layer, which captures nonlinear dependencies among operational variables; and the ontology layer, which enforces semantic and regulatory consistency. These components are coordinated through an orchestration mechanism combining forward analytical processing with backward feedback to achieve convergence under a composite error function.

Experimental validation on wheel-hub crack prediction demonstrated a 56% reduction in predictive error and a 38% improvement in semantic consistency compared with the static reliability baseline. The framework achieved stable convergence within ten orchestration iterations, and the most significant predictors corresponded to known physical degradation factors. These results confirm that statistical, probabilistic, and semantic reasoning can be effectively synthesized into a single computational model.

The proposed approach advances the digital twin concept toward an explainable and certifiable analytical environment for aviation maintenance. By embedding ontological reasoning within the optimization process, the digital twin can verify analytical outputs against engineering knowledge and regulatory logic, thereby supporting traceable and transparent decision-making.

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References

1. Fidanoglu, M. Application of Weibull Distribution Method for Aircraft Part Reliability. *J. Traffic Logist. Eng.* **2017**, *5*, 40–44. [CrossRef]
2. Kontrec, N.; Panić, S. Spare Parts Forecasting Based on Reliability. In *System Reliability*; InTech: London, UK, 2017. [CrossRef]
3. Karaoğlu, U.; Mbah, O.; Zeeshan, Q. Applications of Machine Learning in Aircraft Maintenance. *J. Eng. Manag. Syst. Eng.* **2023**, *2*, 76–95. [CrossRef]
4. Kabashkin, I. Digital Twin Framework for Aircraft Lifecycle Management Based on Data-Driven Models. *Mathematics* **2024**, *12*, 2979. [CrossRef]
5. Elshahhat, A.; Abo-Kasem, O.E.; Mohammed, H.S. Survival Analysis of the PRC Model from Adaptive Progressively Hybrid Type-II Censoring and Its Engineering Applications. *Mathematics* **2023**, *11*, 3124. [CrossRef]
6. Kabashkin, I.; Susanin, V. Decision-Making Model for Life Cycle Management of Aircraft Components. *Mathematics* **2024**, *12*, 3549. [CrossRef]
7. Lázaro, F.L.; Santos, L.F.F.M.; Valério, D.; Melicio, R. Artificial Intelligence and Aviation: A Deep Learning Strategy for Improved Data Classification and Management. *Appl. Sci.* **2025**, *15*, 9403. [CrossRef]
8. Jasra, S.K.; Valentino, G.; Muscat, A.; Camilleri, R. A Comparative Study of Unsupervised Machine Learning Methods for Anomaly Detection in Flight Data: Case Studies from Real-World Flight Operations. *Aerospace* **2025**, *12*, 151. [CrossRef]
9. Payette, M.; Abdul-Nour, G. Machine Learning Applications for Reliability Engineering: A Review. *Sustainability* **2023**, *15*, 6270. [CrossRef]
10. Pendyala, V.; Kim, H. Assessing the Reliability of Machine Learning Models Applied to the Mental Health Domain Using Explainable AI. *Electronics* **2024**, *13*, 1025. [CrossRef]

11. Kabashkin, I. Ontology-Driven Digital Twin Framework for Aviation Maintenance and Operations. *Mathematics* **2025**, *13*, 2817. [CrossRef]
12. Souanef, T.; Al-Rubaye, S.; Tsourdos, A.; Ayo, S.; Panagiotakopoulos, D. Digital Twin Development for the Airspace of the Future. *Drones* **2023**, *7*, 484. [CrossRef]
13. Laukotka, F.N.; Krause, D. Supporting Digital Twins for the Retrofit in Aviation by a Model-Driven Data Handling. *Systems* **2023**, *11*, 142. [CrossRef]
14. Kabashkin, I. Federated Unlearning Framework for Digital Twin–Based Aviation Health Monitoring under Sensor Drift and Data Corruption. *Electronics* **2025**, *14*, 2968. [CrossRef]
15. Botín-Sanabria, D.M.; Mihaita, A.-S.; Peimbert-García, R.E.; Ramírez-Moreno, M.A.; Ramírez-Mendoza, R.A.; Lozoya-Santos, J.J. Digital Twin Technology Challenges and Applications: A Comprehensive Review. *Remote Sens.* **2022**, *14*, 1335. [CrossRef]
16. Wang, T.; Liu, Z. Digital Twin and Its Application for the Maintenance of Aircraft. In *Handbook of Nondestructive Evaluation 4.0*; Meyendorf, N., Ida, N., Singh, R., Vrana, J., Eds.; Springer: Cham, Switzerland, 2021; pp. 1–19. [CrossRef]
17. Moenck, K.H.; Laukotka, F.N.; Deneke, C.; Schüppstuhl, T.; Krause, D.; Nagel, T.J. *Towards an Intelligent Digital Cabin Twin to Support an Aircraft's Retrofit and Base Maintenance*; SAE Technical Papers 2022-01-0046; SAE International: Warrendale, PA, USA, 2022. [CrossRef]
18. Zaccaria, V.; Stenfelt, M.; Aslanidou, I.; Kyprianidis, K.G. Fleet Monitoring and Diagnostics Framework Based on Digital Twin of Aeroengines. In Proceedings of the ASME Turbo Expo, Oslo, Norway, 11–15 June 2018; Volume 6. [CrossRef]
19. Yang, Y.; Ma, M.; Zhou, Z.; Sun, C.; Yan, R. Dynamic Model-Based Digital Twin for Crack Detection of Aeroengine Disk. In Proceedings of the 2021 International Conference on Sensing, Measurement & Data Analytics in the Era of Artificial Intelligence (ICSMD), Chongqing, China, 15–17 October 2021; pp. 1–6. [CrossRef]
20. Wang, T.; Liu, Z.; Liao, M.; Mrad, N. Life Prediction for Aircraft Structure Based on Bayesian Inference: Towards a Digital Twin Ecosystem. In Proceedings of the Annual Conference of the PHM Society, Virtual, 9–13 November 2020; Volume 12, 8p. [CrossRef]
21. Chowdhury, S.H.; Ali, F.; Jennions, I.K. A Methodology for the Experimental Validation of an Aircraft ECS Digital Twin Targeting System Level Diagnostics. In Proceedings of the Annual Conference of the PHM Society, Scottsdale, AZ, USA, 21–26 September 2019.
22. Ezhilarasu, C.M.; Jennions, I.K. A System-Level Failure Propagation Detectability Using ANFIS for an Aircraft Electrical Power System. *Appl. Sci.* **2020**, *10*, 2854. [CrossRef]
23. Ezhilarasu, C.M.; Jennions, I.K. Development and Implementation of a Framework for Aerospace Vehicle Reasoning (FAVER). *IEEE Access* **2021**, *9*, 108028–108048. [CrossRef]
24. Ezhilarasu, C.M.; Skaf, Z.; Jennions, I.K. A Generalised Methodology for the Diagnosis of Aircraft Systems. *IEEE Access* **2021**, *9*, 11437–11454. [CrossRef]
25. Ramesh, G.; Garza, P.; Perinpanayagam, S. Digital Simulation and Identification of Faults with Neural Network Reasoners in Brushed Actuators Employed in an E-Brake System. *Appl. Sci.* **2021**, *11*, 9171. [CrossRef]
26. Huang, Y.; Tao, J.; Sun, G.; Wu, T.; Yu, L.; Zhao, X. A Novel Digital Twin Approach Based on Deep Multimodal Information Fusion for Aero-Engine Fault Diagnosis. *Energy* **2023**, *270*, 126894. [CrossRef]
27. Hazbon Alvarez, O.; Gutierrez Zea, L.; Bil, C.; Napolitano, M.; Fravolini, M.L. Digital Twin Concept for Aircraft Sensor Failure. In *Advances in Transdisciplinary Engineering*; IOS Press: Amsterdam, The Netherlands, 2019; Volume 10, pp. 370–379. [CrossRef]
28. Peng, C.-C.; Chen, Y.-H. Digital Twins-Based Online Monitoring of TFE-731 Turbofan Engine Using Fast Orthogonal Search. *IEEE Syst. J.* **2021**, *16*, 3060–3071. [CrossRef]
29. Kabashkin, I.; Shoshin, L. Artificial Intelligence of Things as New Paradigm in Aviation Health Monitoring Systems. *Future Internet* **2024**, *16*, 276. [CrossRef]
30. Saberi, S.; Kouhizadeh, M.; Sarkis, J.; Shen, L. Blockchain Technology and Its Relationships to Sustainable Supply Chain Management. *Int. J. Prod. Res.* **2019**, *57*, 2117–2135. [CrossRef]
31. Kabashkin, I. The Iceberg Model for Integrated Aircraft Health Monitoring Based on AI, Blockchain, and Data Analytics. *Electronics* **2024**, *13*, 3822. [CrossRef]
32. Chang, H.-M.; Kora, A. The Operation Management Model of Aircraft Maintenance, Repair and Overhaul (MRO) Business. *Int. J. Technol. Eng. Manag. Appl. Sci.* **2014**, *III*, 21–28. Available online: https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=210c90720b9003d165deaeb3c41aed649b32f7ee (accessed on 27 October 2024).
33. Schyga, J.; Hinckeldeyn, J.; Kreutzfeldt, J. Prototype for a Permissioned Blockchain in Aircraft MRO. In Proceedings of the Hamburg International Conference of Logistics (HICL), Hamburg, Germany, 26–27 September 2019; pp. 469–505. [CrossRef]
34. Madhwal, Y.; Panfilov, P.B. Industrial Case: Blockchain on Aircraft's Parts Supply Chain Management. In Proceedings of the American Conference on Information Systems 2017 Workshop on Smart Manufacturing, Boston, MA, USA, 10–12 August 2017; Volume 6. Available online: https://aisel.aisnet.org/sigbd2017/6/ (accessed on 7 October 2024).
35. Sletten, S. Suspected Unapproved Parts in the Aviation Industry: Consideration of System Safety and Control. *J. Aviat. Aerosp. Educ. Res.* **2000**, *9*, 4. Available online: https://pdfs.semanticscholar.org/ab1e/daab31d4a19422a93fe15d3c761fb34d7581.pdf (accessed on 7 October 2024). [CrossRef]

36. Dutta, P.; Choi, T.M.; Somani, S.; Butala, R. Blockchain Technology in Supply Chain Operations: Applications, Challenges, and Research Opportunities. *Transp. Res. Part E Logist. Transp. Rev.* **2020**, *142*, 102067. [CrossRef]
37. Iliuţă, M.-E.; Moisescu, M.-A.; Pop, E.; Ionita, A.-D.; Caramihai, S.-I.; Mitulescu, T.-C. Digital Twin—A Review of the Evolution from Concept to Technology and Its Analytical Perspectives on Applications in Various Fields. *Appl. Sci.* **2024**, *14*, 5454. [CrossRef]
38. Wu, C.-H.; Liu, C.-Y.; Weng, T.-S. Critical Factors and Trends in NFT Technology Innovations. *Sustainability* **2023**, *15*, 7573. [CrossRef]
39. Ekinci, Z. Non-Fungible Tokens and Select Art Law Considerations. *Arts* **2023**, *12*, 192. [CrossRef]
40. López-Pimentel, J.C.; Morales-Rosales, L.A.; Algredo-Badillo, I.; Del-Valle-Soto, C. NFT-Vehicle: A Blockchain-Based Tokenization Architecture to Register Transactions over a Vehicle's Life Cycle. *Mathematics* **2023**, *11*, 2801. [CrossRef]
41. Alketbi, M.A.; Dalalah, D.; Dweiri, F. A Framework for the Characterization of Aviation Construction Projects: The Case of UAE. *Buildings* **2024**, *14*, 2384. [CrossRef]
42. Vivalda, P.; Fioriti, M. Stream Life Cycle Assessment Model for Aircraft Preliminary Design. *Aerospace* **2024**, *11*, 113. [CrossRef]
43. Ershov, M.A.; Klimov, N.A.; Savelenko, V.D.; Makhova, U.A.; Burov, N.O.; Karpunin-Ozherovskiy, E.V.; Aleksanyan, D.R.; Donskaya, E.S.; Mukhina, D.Y.; Kapustin, V.M.; et al. An Overview of the Global Market, Fleet, and Components in the Field of Aviation Gasoline. *Aerospace* **2023**, *10*, 863. [CrossRef]
44. Basora, L.; Bry, P.; Olive, X.; Freeman, F. Aircraft Fleet Health Monitoring with Anomaly Detection Techniques. *Aerospace* **2021**, *8*, 103. [CrossRef]
45. Werbińska-Wojciechowska, S.; Giel, R.; Winiarska, K. Digital Twin Approach for Operation and Maintenance of Transportation System—Systematic Review. *Sensors* **2024**, *24*, 6069. [CrossRef]
46. Rahmani, R.; Jesus, C.; Lopes, S.I. Implementations of Digital Transformation and Digital Twins: Exploring the Factory of the Future. *Processes* **2024**, *12*, 787. [CrossRef]
47. Sepasgozar, S.M.E. Differentiating Digital Twin from Digital Shadow: Elucidating a Paradigm Shift to Expedite a Smart, Sustainable Built Environment. *Buildings* **2021**, *11*, 151. [CrossRef]
48. Frick, N.; Terwolbeck, J.; Seibel, B.; Metternich, J. Design Model for the Digital Shadow of a Value Stream. *Systems* **2024**, *12*, 20. [CrossRef]
49. Mourtzis, D.; Balkamos, N. Design of Manufacturing Systems Based on Digital Shadow and Robust Engineering. *Appl. Sci.* **2023**, *13*, 5184. [CrossRef]
50. Torrejón, A.; Zapata, N.; Bonilla, L.; Bustos, P.; Núñez, P. Design and Development of Shadow: A Cost-Effective Mobile Social Robot for Human-Following Applications. *Electronics* **2024**, *13*, 3444. [CrossRef]
51. Yang, J.; Zheng, Y.; Wu, J.; Wang, Y.; He, J.; Tang, L. Enhancing Manufacturing Excellence with Digital-Twin-Enabled Operational Monitoring and Intelligent Scheduling. *Appl. Sci.* **2024**, *14*, 6622. [CrossRef]
52. Adibi, S.; Rajabifard, A.; Shojaei, D.; Wickramasinghe, N. Enhancing Healthcare through Sensor-Enabled Digital Twins in Smart Environments: A Comprehensive Analysis. *Sensors* **2024**, *24*, 2793. [CrossRef] [PubMed]
53. *ARP4754B*; Guidelines for Development of Civil Aircraft and Systems. SAE International: Warrendale, PA, USA, 2023. [CrossRef]
54. *IEC 60300-3-5*; Dependability Management—Part 3-5: Application Guide—Reliability Test Conditions and Statistical Test Principles. International Electrotechnical Commission (IEC): Geneva, Switzerland, 2001.
55. *IEEE 1413-2010*; IEEE Standard Framework for Reliability Prediction of Hardware. IEEE Reliability Society Standards Committee: New York, NY, USA, 2010.
56. Abernethy, R.B. *The New Weibull Handbook*, 5th ed.; Fulton Findings: North Palm Beach, FL, USA, 2006.
57. Hitzler, P.; Krötzsch, M.; Rudolph, S. *Foundations of Semantic Web Technologies*, 1st ed.; Chapman and Hall/CRC: Boca Raton, FL, USA, 2009. [CrossRef]
58. Zhao, X.; Li, X.; Sun, S.; Jia, X. Secure and Efficient Federated Gradient Boosting Decision Trees. *Appl. Sci.* **2023**, *13*, 4283. [CrossRef]
59. Maharela, I.A.; Fletcher, L.; Chen, D.-G. Modified Cox Models: A Simulation Study on Different Survival Distributions, Censoring Rates, and Sample Sizes. *Mathematics* **2024**, *12*, 2903. [CrossRef]

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